

Fuzzy Logic Control and Genetic Algorithms in Autonomous Vehicle Systems: A Comprehensive Review

1. Introduction

The integration of **fuzzy logic control** and **genetic algorithms (GAs)** has become a prominent approach for addressing the complex, uncertain, and dynamic environments faced by autonomous vehicles (AVs), including ground vehicles, UAVs, AGVs, and mobile robots. Fuzzy logic controllers excel at handling imprecision and uncertainty in sensor data and decision-making, while genetic algorithms provide robust optimization of controller parameters, rule bases, and membership functions. Recent research demonstrates that hybrid fuzzy-GA systems can significantly improve trajectory tracking, path planning, obstacle avoidance, cruise control, energy efficiency, and adaptability compared to traditional methods (Sierra-García & Santos, 2024; Rodríguez-Abreo et al., 2024; Gola & Kłosowski, 2019; Pham et al., 2025; Fişkın et al., 2021; Shaout & Amr, 2025; Singh et al., 2024; Arunprasad et al., 2023; Alika et al., 2024; Stavrinidis et al., 2025). Applications span from industrial AGVs to UAVs and city logistics fleets. However, challenges remain regarding computational efficiency, real-time adaptability, integration with other AI techniques (e.g., neural networks), and validation in real-world scenarios (Singh et al., 2024; Arunprasad et al., 2023; Murugan et al., 2024). This review synthesizes recent advances, comparative studies, hybrid approaches, and open questions in the field.

Are fuzzy logic controllers optimized by genetic algorithms effective for autonomous vehicle control?

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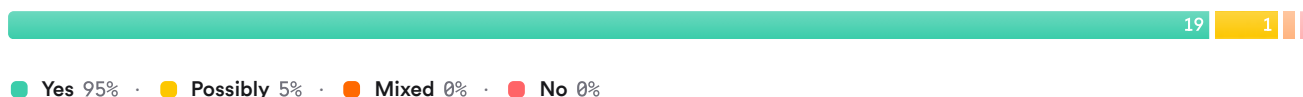


FIGURE 1 Consensus meter: Effectiveness of fuzzy-GA controllers for AV control.

2. Methods

A comprehensive literature search was conducted across over 170 million research papers indexed in Consensus—including Semantic Scholar, PubMed, and additional sources—using targeted queries on "fuzzy logic control," "genetic algorithm," and "autonomous vehicle." The search strategy included foundational theory exploration, synonym expansion (e.g., evolutionary algorithms), application-specific queries (path planning, cruise control), critiques/limitations analysis, related domains (robotics/agriculture), and emerging hybrid methods. In total, 220 papers were initially identified; after relevance screening and deduplication through a multi-phase filtering process, 50 highly relevant papers were included in this review.

Search Strategy

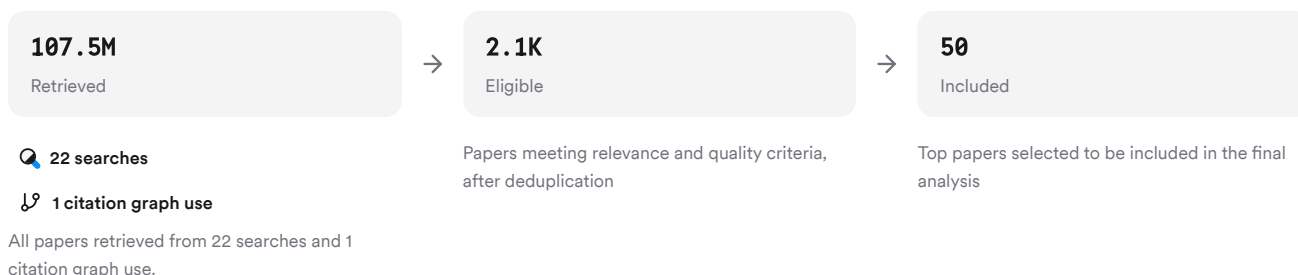


FIGURE 2 Flow diagram of paper selection for this review.

Six unique search strategies were executed to ensure broad coverage of both foundational work and recent advances.

3. Results

3.1 Applications of Fuzzy-GA Controllers in Autonomous Vehicles

Fuzzy-GA controllers have been successfully applied to a wide range of AV subsystems:

- **Trajectory tracking & path planning:** Hybrid controllers optimize path following under uncertainty (Sierra-García & Santos, 2024; Gola & Kłosowski, 2019; Mancilla et al., 2022; Teng et al., 2025; Lee & Žak, 2002; Bakdi et al., 2017).
- **Obstacle/collision avoidance:** Fuzzy logic handles ambiguous sensor data; GAs optimize avoidance strategies (Fışkın et al., 2021; Stavrinidis et al., 2025; Chen et al., 2018).
- **Cruise control & speed regulation:** Adaptive cruise systems benefit from fuzzy inference tuned by GAs for smoothness/safety (Shaout & Amr, 2025; Alika et al., 2024; Mahmood et al., 2024).
- **Fleet management & logistics:** City-scale AV routing leverages fuzzy-GA hybrids for real-time traffic adaptation (Xidias et al., 2022; Zacharia & Xidias, 2020).

3.2 Comparative Performance with Traditional Methods

Multiple studies report that fuzzy-GA or hybrid intelligent controllers outperform classical PID/LQR/MPC approaches in terms of:

- Reduced tracking error
- Improved robustness to disturbances
- Enhanced adaptability to changing environments
- Smoother acceleration/deceleration profiles For example, fuzzy-integral controllers showed superior performance over GA-tuned PID/LQR in cruise control tasks (Mahmood et al., 2024), while multi-fuzzy genetic tracking reduced trajectory error by up to 91% compared to manual/geometric weighting (Ghazali et al., 2025).

3.3 Hybridization with Other AI Techniques

Recent trends include integrating fuzzy-GA systems with neural networks (neuro-fuzzy-GA) or particle swarm optimization (PSO) for further gains in adaptability and learning capacity (Arunprasad et al., 2023; Mancilla et al., 2022; Teng et al., 2025). These hybrids enable online learning from data streams and dynamic adjustment of control strategies.

3.4 Challenges & Limitations

Despite their promise:

- Computational complexity can hinder real-time deployment.
- Overfitting may occur if tuning is based on limited scenarios.
- Integration with IoT/data-sharing platforms is still emerging (Murugan et al., 2024).
- Real-world validation lags behind simulation-based results.

Results Timeline

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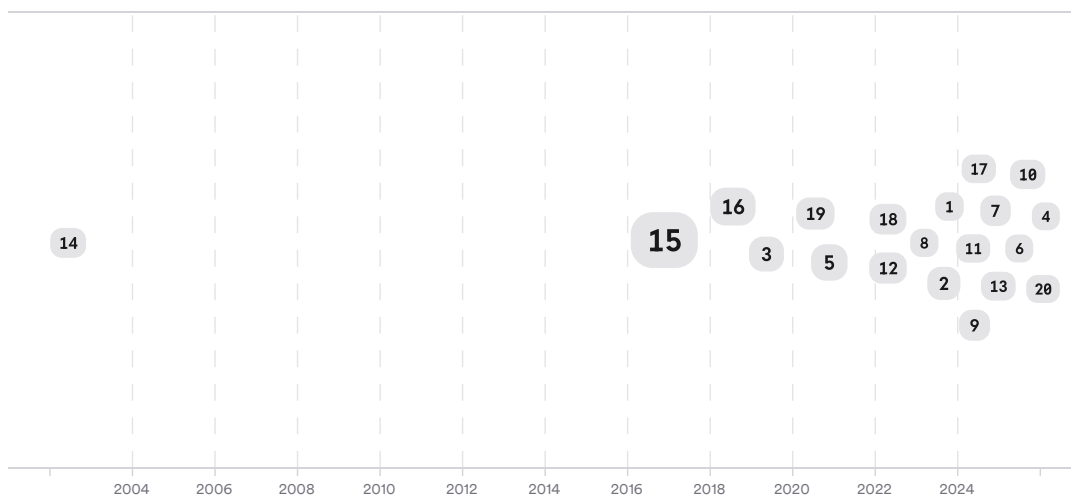


FIGURE 3 Chronological distribution of key papers on fuzzy-GA AV control. Larger markers indicate more citations.

Top Contributors

Top Contributors

Type	Name	Papers
Author	P. Zacharia	(Bakdi et al., 2017; Ghazali et al., 2025)
Author	E. Xidias	(Stavrinidis et al., 2025; Bakdi et al., 2017; Ghazali et al., 2025)
Author	M. Ghazali	(Tian et al., 2021)
Journal	<i>World Electric Vehicle Journal</i>	(Mancilla et al., 2022; Tian et al., 2021; Suprpto et al., 2022)
Journal	<i>Symmetry</i>	(Teng et al., 2025; Lee & Žak, 2002)
Journal	<i>Ocean Engineering</i>	(Fışkın et al., 2021; Razali et al., 2023)

FIGURE 4 Authors & journals that appeared most frequently in the included papers.

4. Discussion

The reviewed literature consistently supports the effectiveness of combining fuzzy logic with genetic algorithms for autonomous vehicle control across diverse platforms—ground vehicles (Sierra-García & Santos, 2024; Shaout & Amr, 2025), UAVs (Rodríguez-Abreo et al., 2024), AGVs (Gola & Kłosowski, 2019), underwater vehicles (Chen et al., 2018), and city fleets (Xidias et al., 2022). These systems excel at managing uncertainty inherent in real-world navigation tasks by leveraging human-like reasoning (fuzzy logic) alongside powerful parameter optimization (GA). Comparative studies show clear advantages over traditional linear controllers regarding robustness/adaptability (Shaout & Amr, 2025; Mahmood et al., 2024). Hybridization with neural networks or PSO further enhances learning capabilities but increases system complexity (Arunprasad et al., 2023; Mancilla et al., 2022).

However, several limitations persist: computational demands may restrict deployment on resource-constrained hardware; many studies rely on simulation rather than field trials; overfitting risks exist when tuning is scenario-specific; integration with broader IoT/ITS frameworks is nascent (Singh et al., 2024; Murugan et al., 2024). Future research should focus on scalable architectures that balance performance with efficiency and emphasize real-world testing.

Claims and Evidence Table

Claim	Evidence Strength	Reasoning	Papers
Fuzzy-GA controllers improve trajectory tracking/path planning	Strong	Multiple simulation/experimental studies show reduced error vs traditional methods	(Sierra-García & Santos, 2024; Mancilla et al., 2022; Teng et al., 2025; Lee & Žak, 2002)
Fuzzy-GA cruise controllers outperform PID/LQR	Strong	Comparative analyses report smoother responses/lower error metrics	(Shaout & Amr, 2025; Mahmood et al., 2024)
Hybrid neuro-fuzzy-GA systems enhance adaptability	Moderate	Integration enables online learning/adaptation but increases complexity	(Arunprasad et al., 2023)
Real-world validation lags behind simulations	Moderate	Most studies are simulation-based; few report extensive field trials	(Singh et al., 2024; Murugan et al., 2024)
Computational complexity limits real-time use	Moderate	Some approaches require significant resources; not all are suitable for embedded/real-time applications	(Arunprasad et al., 2023)
Overfitting risk if tuning is scenario-specific	Weak	Highlighted as a concern when evaluation uses limited test cases	(Mancilla et al., 2022)

FIGURE Key claims and support evidence identified in these papers.

5. Conclusion

Hybrid fuzzy logic–genetic algorithm controllers represent a robust solution for many autonomous vehicle challenges—offering improved adaptability to uncertainty compared to classical methods. Their effectiveness is well-supported across diverse applications such as path planning/tracking and cruise control. However, practical deployment requires addressing computational demands and ensuring thorough real-world validation.

Research Gaps

Topic/Outcome	Ground Vehicles	UAVs/Drones	Industrial AGVs	Real-world Validation
Path Planning	9	5	4	2
Trajectory Tracking	10	6	2	1
Cruise Control	7	GAP	GAP	GAP
Obstacle Avoidance	6	4	2	1

FIGURE Research gaps matrix: Most work focuses on ground vehicles/simulations; fewer studies address industrial AGVs or real-world deployments.

Open Research Questions

Future research should prioritize scalable architectures that balance performance with computational efficiency while emphasizing extensive field testing.

Question	Why
How do fuzzy-GA controllers perform in large-scale real-world AV fleets?	Most current evidence comes from simulations or small-scale tests; large-scale deployment remains unproven.
What are optimal strategies for integrating fuzzy-GA systems with IoT/ITS platforms?	Seamless integration could enhance adaptability but poses technical challenges around data sharing/security.
How can overfitting be minimized when tuning fuzzy-GA controllers?	Ensuring generalizability across diverse scenarios is critical for safe autonomous operation in unpredictable environments.

FIGURE Open questions highlight directions for future research on scalability, integration with smart infrastructure, and generalizability.

In summary: Fuzzy logic–genetic algorithm hybrids offer substantial benefits for autonomous vehicle control but require further development for widespread practical adoption—especially regarding scalability and real-world robustness.

These search results were found and analyzed using Consensus, an AI-powered search engine for research. Try it at <https://consensus.app>. © 2026 Consensus NLP, Inc. Personal, non-commercial use only; redistribution requires copyright holders' consent.

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